



Error Makes Clever

Backend Development Roadmap





Backend Framework

- Choose one language and its framework:



Django: building full-stack applications.
Flask: lightweight, flexible projects.



Spring Boot: simplifies creating enterprise -level applications.



Node.js + Express.js: fast, scalable server-side apps





Version Control

- Learn **Git basics** like cloning, branching, and merging on platforms like GitHub or GitLab.
- Practice **resolving merge conflicts** to handle collaborative coding effectively.



Cloning

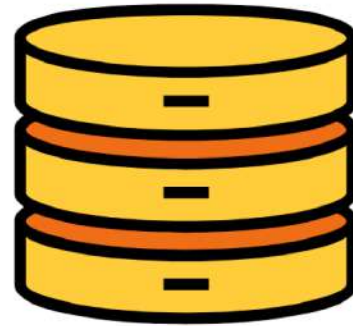
Branching

Merging





Databases



Relational

NoSQL

MySQL, PostgreSQL
for structured data.

MongoDB for flexible,
document-based
data.

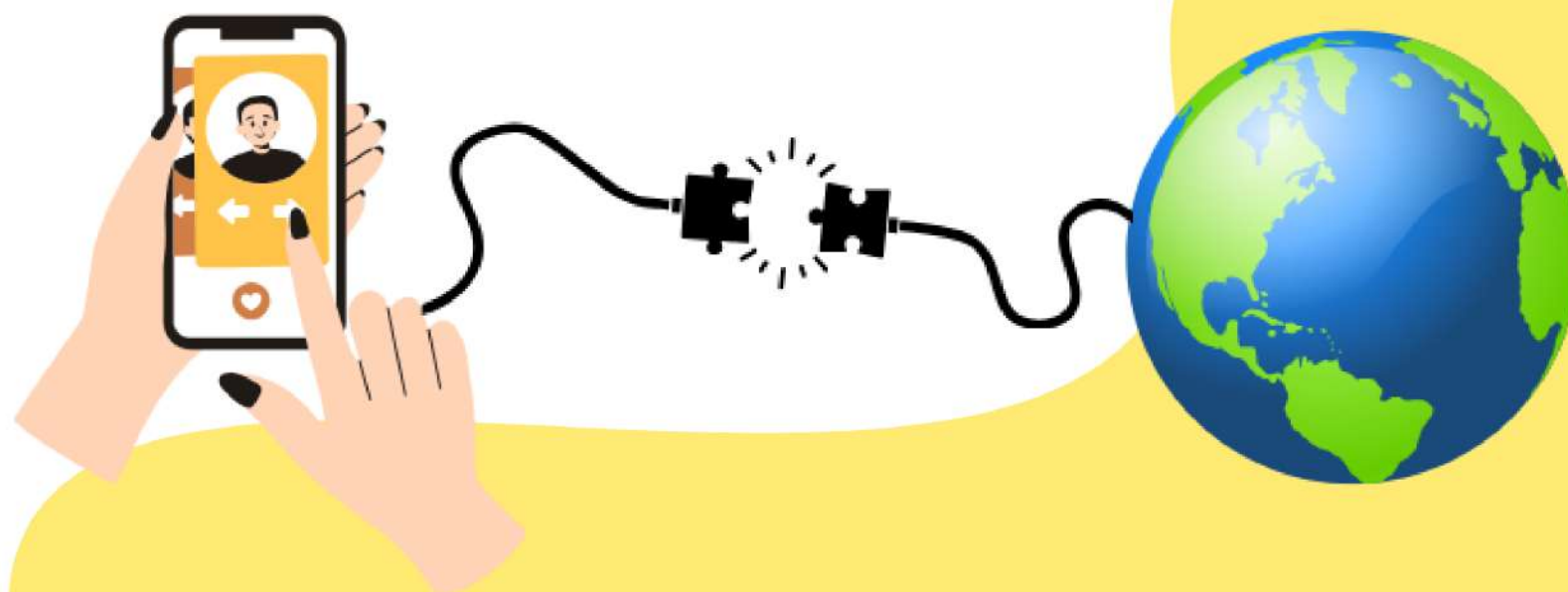
- Practice schema design, normalization, and query writing.





APIs and Communication

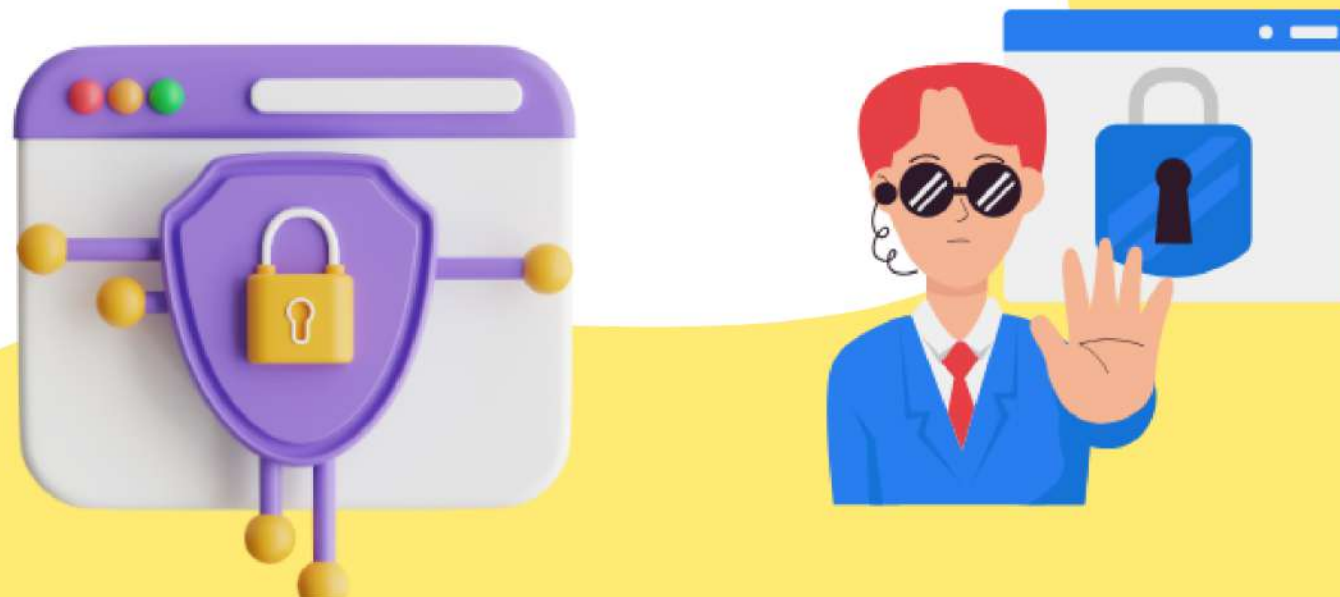
- APIs connect your app to the world.
- Learn REST **APIs**, **JSON**, and **HTTP** methods, and test APIs with tools like Postman.





Web Security

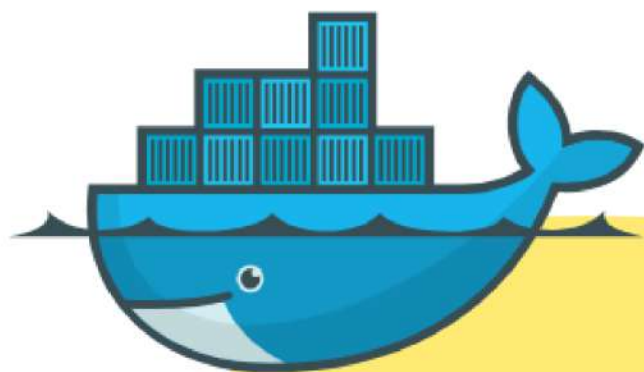
- Keep your app safe by **implementing authentication** (OAuth2, JWT), **preventing vulnerabilities** (SQL Injection, XSS, CSRF), and **using HTTPS/SSL** for secure connections.





Containerization and Deployment

- Containerization: Use **Docker** to package your app.
- Deployment: Host your app on Heroku, AWS, or Render.
- Learn **CI/CD pipelines** to automate updates.





Advanced Topics

- To Take your skills further, explore
- **Web Servers:** Dive into Nginx, Apache, and load balancing.
- **Testing:** Write unit and integration tests to ensure reliability.
- **Caching:** Speed up apps with Redis or Memcached.





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Kickstart your backend development journey

save this roadmap and build a strong foundation for a
successful career effortlessly!



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